The smallest critical sizes are obtained for *homogeneous systems* of pure fissile nuclides with maximum neutron reflection. For neutrons with the fission energy spectrum, the critical mass of a metallic sphere of pure ^{235}U is 22.8 kg, that of ^{233}U is 7.5 kg, and that of ^{239}Pu is 5.6 kg, assuming a 20 cm uranium metal neutron reflector. For fission by thermal neutrons the smallest critical size of a spherical *homogeneous aqueous solution* of $^{235}UO_2SO_4$ without reflector requires 0.82 kg of ^{235}U in 6.3 l of solution. The corresponding figures for ^{233}U are 0.59 kg in 3.3 l, and of ^{239}Pu, 0.51 kg in 4.5 l.

Homogeneous solutions of fissile nuclides are produced in the reprocessing of spent fuel elements, where care must be exercised that the critical size is not exceeded in any equipment or container in order to prevent an accidental chain reaction. Several such accidents have occurred in the past in fuel production and reprocessing plants in which very high doses were received by nearby personnel even though the duration of the chain reaction usually was very short and a violent explosion never occurred.

For heterogeneous reactors it is more difficult to quote comparable simple values for critical size. These have to be calculated by numerical methods or determined empirically for each particular reactor configuration.

19.8. REACTOR KINETICS

The mean lifetime θ for a neutron in a reactor is the time it takes on the average for the neutrons to complete one loop in the neutron cycle. In thermal reactors θ is 10^{-3}–10^{-4} s due to the comparatively low speed of thermalized neutrons and average distance in the moderator covered by random walk when travelling from and to a fuel pin. For each loop the number of neutrons is multiplied by a factor k_{eff}. Since one neutron is used for maintaining the chain reaction, the neutrons in the reactor change with time according to

$$\mathrm{d}N/\mathrm{d}t = N\left(k_{\text{eff}} - 1\right)/\theta + K \tag{19.26}$$

where N is the total number of neutrons in the reactor, and K is the contribution from any constant neutron source present. Solving this equation, we obtain

$$N = N_0\, \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta} + K\,\theta\left(1 - \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta}\right)\Big/\left(1 - k_{\text{eff}}\right) \tag{19.27}$$

where N_0 is the number of neutrons at $t = 0$. We can distinguish three different cases:

(i) $k_{\text{eff}} < 1$ and $t \gg \theta$: the reactor is subcritical,

$$N(t) = K\,\theta\left(1 - k_{\text{eff}}\right)^{-1} \tag{19.28}$$

The initial number of neutrons, N_0, has disappeared and the number of neutrons is directly related to the constant neutron source. The reactor acts as a neutron amplifier, with amplification increasing with k_{eff}.

(ii) $k_{\text{eff}} > 1$; the reactor is supercritical,

$$N(t) = \left[N_0 - K\,\theta/\left(k_{\text{eff}} - 1\right)\right] \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta} + K\,\theta/\left(k_{\text{eff}} - 1\right) \tag{19.29}$$

If K is so small that the terms containing $K\theta$ can be neglected this reduces to

$$N(t) = N_0\, \mathrm{e}^{\left(k_{\text{eff}}-1\right)t/\theta} \tag{19.30}$$

The neutron flux (and consequently also power) increases exponentially. All reactors have $K > 0$, the reason being neutron production through spontaneous fission in ^{238}U and other actinides or through other nuclear reactions. However, the resulting neutron source strength is usually not large enough to give a reliable indication on the control instruments of the power level of a new reactor in the initial start-up procedure. Therefore, extra neutron sources are generally introduced in reactors to facilitate starting. A common type is Sb+Be, where ^{124}Sb (half-life 60 d) has been produced through irradiation in a reactor using the (n,γ) reaction in ^{123}Sb (43% abundance). The neutrons are emitted from (γ,n) reactions in Be. Also ^{241}Am+Be sources are used. When the reactor has reached desired power, control rods are inserted, which decrease k_{eff} to 1. Later the partially spent fuel contains sufficient amounts of nuclides with spontaneous fission, e.g. ^{252}Cf, to make a separate neutron source superfluous.

(iii) $k_{\text{eff}} = 1$; the reaction is just critical,

$$N(t) = N_0 + K\,t \tag{19.31}$$

Although this equation indicates that the number of neutrons would increase slightly with time, the $K\,t$ term is usually negligible.

In reactor technology it is common to speak of the *reactivity* ρ and *excess reactivity* (Δk), which are defined by the expression

$$\Delta k \equiv k_{\text{eff}} - 1 = \rho\, k_{\text{eff}} \tag{19.32}$$

Since k_{eff} is close to 1 in a properly operating reactor, we have $\rho \approx \Delta k$. The *reactor time constant* (or *period*) is defined as

$$t_{\text{per}} = \theta/\rho \approx \theta/\Delta k = \theta/\left(k_{\text{eff}} - 1\right) \tag{19.33}$$

As the neutron flux increases in a supercritical reactor, the second term in (19.29) becomes negligible compared to N_0 and the equation can be simplified to

$$N = N_0\, \mathrm{e}^{\left(t/t_{\text{per}}\right)} \tag{19.34}$$

The shorter t_{per} is the faster is the increase in the neutron flux (and reactor power). With $\theta = 10^{-3}$ and $k_{\text{eff}} = 1.1$, Δk is 0.1 and $t_{\text{per}}\ 10^{-2}$. For a longer t_{per} of 1 s, the number of neutrons would increase with a factor of $\mathrm{e}^{10} = 10^4$ every 10 s, which still is much too

rapid for safe and simple control of a reactor. As described below, nuclear reactors are designed to avoid this problem.

In the equations above, θ is the mean lifetime for the prompt neutrons. As discussed in §11.7.1, some fission products decay by β^- leading to an excited state which emits a neutron, *β^--delayed neutron emission*. For example ^{87}Kr decays partly by β^--delayed neutron emission; it is a daughter of ^{87}Br with a half-life as long as 56 s. A large number of such neutron emitting fission products have been discovered, all with shorter half-lives, see Table 19.5. In reactors where the moderator contains D or Be atoms, γ,n reactions with energetic γ-rays from shortlived fission products and from activation products is also a source of delayed neutrons. The delayed neutrons have lower kinetic energies ($\sim$0.5 MeV) than the prompt ones and amount to <1% of the total number of fission neutrons emitted: the fraction of delayed neutrons, β, is 0.27% for ^{233}U, 0.65% for ^{235}U, 0.21% for ^{239}Pu, and 0.52% for ^{241}Pu. When the delayed neutrons are taken into account, the *effective neutron generation time* is, approximately,

$$\theta_{\text{eff}} = \theta + \Sigma(\beta_i/\lambda_i) \tag{19.35}$$

where β_i is the fraction of fission neutrons which are delayed with a decay constant λ_i. The summation gives the mean lifetime of the delayed neutrons; for ^{233}U 0.049 s, ^{235}U 0.084 s, ^{239}Pu 0.033 s, and ^{241}Pu $\sim$0.06 s. Thus for ^{235}U $\theta_{\text{eff}} \approx 10^{-3} + 0.084$. Because the generation time determines the period of the reactor the delayed neutrons have lengthened the reactor period by almost a factor of 100, thereby making reactor control much more manageable.

t_{per} depends on Δk in such a way that if $\Delta k > \beta$ the delayed neutrons are not able to make their decisive influence on the reactor period, and the rate of neutron production is

Table 19.5 Some important fission products partly decaying by β-delayed n-emission

Isotope	Half-life (s)	decay by β-n (%)	Cumulative fission yield (%)			
			^{233}U	^{235}U	^{239}Pu	^{241}Pu
^{85}As	2.03	23.0	0.127	0.200	0.061	0.084
^{87}Br	55.7	2.6	2.67	1.82	0.608	0.633
^{88}Br	16.7	6.6	1.90	1.68	0.645	0.617
^{89}Br	4.37	14.2	0.677	1.64	0.346	0.456
^{90}Br	1.92	24.6	0.275	0.620	0.163	0.249
^{93}Rb	5.85	1.4	2.28	2.94	1.71	1.80
^{94}Rb	2.73	10.4	0.849	1.54	0.754	1.22
^{98}Y	2.0	3.4	1.64	2.99	2.59	3.58
^{135}Sb	1.71	17.5	0.031	0.165	0.006	0.490
^{136}Te	17.5	0.7	0.005	2.57	0.005	1.98
^{137}I	24.5	6.4	1.35	2.60	2.46	4.56
^{138}I	6.4	5.5	0.828	1.06	0.436	2.60
^{139}I	2.29	9.5	0.171	0.119	0.376	1.16

dependent only on the prompt neutrons. Such a reactor is said to be *prompt critical*. Nuclear explosives are designed to be prompt critical with a $\Delta k \geq 1$ giving a $t_{per} \leq 10^{-8}$ s, while reactors for power production (thermal as well as fast reactors) should for safety reasons always be operated in the *delayed critical* region. If in a thermal uranium reactor Δk is made $\geq \beta$ ($k_{eff} \geq 1.0065$), t_{per} becomes 13 s for $\theta_{eff} = 0.084$, and (19.34) shows that the neutron amount (and power) doubles in about 10 s. Usually k_{eff} is made smaller than 1.0065 and the power doubling time is correspondingly larger.

Some of the isobar chains from fission contain isotopes with very high n,γ cross sections. One of the most important is $A = 135$ and it's effect on reactor kinetics is used as an example. The genetic relations in the A 135 chain is:

$$\begin{array}{lccccccc} \text{fission yield\%} & 3.5 & & 2.5 & & 0.6 & & \\ & \downarrow & & \downarrow & & \downarrow & & \\ & {}^{135}\text{Te} & \xrightarrow{18\text{s}} & {}^{135}\text{I} & \xrightarrow{6.61\text{ h}} & {}^{135}\text{Xe} & \xrightarrow{9.10\text{ h}} & {}^{135}\text{Cs} \xrightarrow{2\text{ My}} {}^{135}\text{Ba(stable)} \\ & & & & & \text{n},\gamma \downarrow & 2.65\times10^{6}\text{ b} & \\ & & & & & {}^{136}\text{Xe(stable)} & & \\ & & & & & \text{n},\gamma \downarrow & 0.16\text{ b} & \end{array} \tag{19.36}$$

During operation at a constant neutron flux the large $\sigma_{n,\gamma}$ continuously converts ^{135}Xe to ^{136}Xe, thereby keeping the amount small and limiting the poisoning effect. However, after a strong reduction in neutron flux ^{135}Xe formation is more rapid by decay of its precursor, ^{135}I, than its destruction, resulting in a transient radioactive equilibrium. As a result the reactivity drops for some hours because of an increasing amount of ^{135}Xe. If the reactor has been shut down for a while, it may be impossible to start it until the amount of ^{135}Xe has been reduced again by decay (*xenon poisoning*). However, if the reactor can be started it burns ^{135}Xe and Δk increases rapidly until a new equilibrium concentration is reached. This is called a *xenon transient* and can temporarily make the reactor difficult to control; if mishandled it may even cause prompt criticality. Once xenon poisoning has become very important after a reactor stop it is practice to delay restart until ^{135}Xe has decreased to a safe level.

19.9. FUEL UTILIZATION

Figure 19.7 shows the consumption of fissile ^{235}U while new fissile ^{239}Pu and ^{241}Pu (as well as some fission products and other actinides) are produced through radiative capture in fertile ^{238}U and ^{240}Pu. The Figure relates to a particular reactor type, and different reactors give somewhat different curves. Figure 19.5 shows the different capture and decay reactions. In §19.2 it is concluded that in a reactor operating at a power of 3 GW_{th} 3.16 kg ^{235}U d^{-1} would be fissioned. Because of the ^{235}U(n,γ)^{236}U reaction, some additional uranium is consumed (19.4). To account for this, we use